# Observer Directed Native Qualification

Mettapedia Working Notes

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**Abstract**

A deterministic source generator crosses thirteen candidate native execution families with six explicit observers. Its runner distinguishes answer bags, ordered values, caught errors and effect traces, preserves reproducible execution receipts, and shrinks confirmed discrepancies under declared reductions. Scalar admission controls are exported to the existing Lean source extractor. The original r82 cohort agrees on its first twelve families; an added existence family exposes false absence for open compact facts. A separately frozen successor candidate passes the expanded fourteen-route handoff. Hash-pinned debugger probes distinguish worker completion from mere recognizer entry, and archived mutants qualify the generator's boundary cases. This is empirical qualification infrastructure, not a universal correctness proof. Separate S6 modules establish the ownership and transfer correspondence laws.

## 1 The observer is part of the test

Write $O_C(P)$ for the observation of program $P$ under a declared contract $C$. The native/reference obligation is

$$O_C(P_{\text{native}}) = O_C(P_{\text{ordinary}}).$$

It is not equality after an arbitrary common normalization. In particular, seeing parentheses in an output does not license treating its fields as an unordered bag.

The generated source wraps query results in fresh, numbered observation packets. Packet interpretation comes from the case manifest:

| Contract | Preserved distinctions |
|---|---|
| Answer bag | Answer multiplicity, term structure and shared-variable aliasing |
| Collected bag | Element multiplicity and ordered fields within each element |
| Ordered value | Element and field order, literal strings and term structure |
| Caught error | The actual error packet, not just a success/failure bit |
| Effect trace | Exact order and text, including whitespace and line endings |

Alpha comparison uses the existing complete-collection bijection, so two answers sharing a variable cannot be renamed independently. Literal strings containing variable-looking text remain literal strings. Unmarked output is compared as an ordered text trace. Only explicitly requested runtime counters and dense-storage diagnostics are separated from stderr; other stderr remains an observation.

Timeout, output-limit termination, invalid UTF-8, a native crash, a failed reference program and malformed observation packets have distinct recorded outcomes. They never become matching empty answer bags. Output capture is bounded, and a stopped test terminates its own process group.

## 2 Source generation and admission

The candidate families are scalar folds, filtered matches, grouped counts, listed matches, dead rules, kept relations, row take-over, open joins, guarded insert batches, arithmetic folds, saturation, relaxation and ground-pattern match existence. Each family records its inspected source entry point. This inventory covers the native query/fold/dense-region lane; it is not an inventory of every optimization in the runtime's other language engines. The HE coordinate-form existence caller is a fourteenth, separately driven route because it has a different language surface and a partitioned compact/row contract. The machine-readable inventory distinguishes named function returns, inlined checkpoints, operation traces and this caller chain.

The scalar syntax mirrors `Machines.TypedScalar`'s data/test distinction. Constructors reject an arithmetic operation on a predicate, except for an explicit negative-control node that requests ordinary fallback. The source contains signed integers, mixed Boolean/numeric/symbol keys, floats in relational controls, repeated variables, duplicate occurrences and selected integer-overflow and non-integer-order cases.

The generator's Python checks are not a proof of the Python implementation. It exports each scalar body's source admission assertion into a Lean file using the existing `PeTTa.Bridges.ScalarSource.elaborate`. The checked cohort contains twenty-one such assertions, including the rejected Boolean arithmetic control. These assertions concern source extraction, not C execution or whole-fold semantics.

Entry shapes matter independently of output correctness. The saturation schema seeds its insertion function through a match; relaxation uses a listed frontier; grouped counts include the disjoint additions their recognizer requires; arithmetic folds use `foldall`. A matching result from a shape that always declines is not evidence for the intended specialization.

Collective consumers are generated separately from streaming contexts. For each relevant query family, count and existence results flow to all six observers. Open-join controls include stored variables and set effects; listed-match controls include flat facts, duplicate list elements and at least sixteen entries. The nested, streaming and mixed-kind cases remain in the corpus: their ordinary fallback is tested rather than silently reclassified as worker coverage.

## 3 Metamorphic laws need hypotheses

The runner tests an additional unrelated equation against the original program, independently on both execution routes. It also compares observations across plain-row, normal dense and eager dense storage. The source snapshots, binary hash, environment switches and case parameters are retained.

Two controls prevent unlicensed ordering assumptions. First, numerically equal but differently represented values can tie under sorting. Ordered and effect probes therefore construct ground structural keys using `repr`, sort those keys with their original payloads, and retain the payloads for the observer. Second, the arithmetic-fold input is an unordered match; its positive step is additive in the accumulator. Explicit-list scalar folds remain available for order-sensitive steps.

Observer construction itself has source semantics. Generated binders are fresh relative to the region. A computed collection is bound before it is passed to `superpose`; placing a sorting computation directly in that raw argument position would enumerate syntax rather than the intended values. Twelve fixed controls check the complete observer outputs for integer 7 and Boolean true, including the stored effect values. Nonempty witnesses additionally prevent an accidentally unproductive positive schema from passing vacuously.

These checks do not justify arbitrary input permutations or arbitrary changes to function def-

initions. Each added metamorphic transformation needs its own source-level preservation conditions.

## 4 Reproduction and shrinking

Every case is serializable and can be replayed independently. The output directory records the generated source, binary and harness hashes, observer contracts, route settings, stdout, stderr, exit disposition, counters and comparison verdicts. Source-file hashes are an inspected working-source snapshot; correspondence to a frozen binary still requires its build receipt.

The shrinker removes inputs, simplifies expression subtrees and integer literals, reduces storage histories and removes optional boundary features. It preserves family-specific conditions such as the additive arithmetic-fold schema. A candidate must reproduce the same family/observer discrepancy twice. Where direct route evidence exists, the candidate must retain that evidence too. Equal source programs are not redundantly tested merely because unused generator metadata changed.

The result is either a local fixed point under the declared reductions, a budget-limited result or an unstable/nonreproducible case. No global-minimum claim is made. On the frozen Boolean-bug baseline, a generated nested equality reduced to a one-item equality fold; the saved case replays without a discrepancy on the repaired binary.

## 5 Direct evidence from a frozen executable

The reviewed probe profile names the r82 executable by its full SHA-256 and requires its x86-64 ABI. Unknown executable hashes have no admitted profile. Named workers are observed at instruction entry and completion; successful return tags are distinguished from refusal and interruption. Inlined grouped, listed and saturation paths use reviewed instruction checkpoints relative to their containing symbol. A checkpoint is associated with its enclosing invocation, whose completion is checked too. Dead-rule pruning and saturation success must belong to the same invocation.

A generic native-query completion is not credited to a guarded batch unless the query actually has an effect consumer, effects and a guard. Its field offsets and return tags were checked against compiled headers and accesses in the pinned executable. Kept-storage activation can occur on both query routes: the query-reference switch does not disable relation representation.

The route-coverage debugger calls no inferior functions and writes no data or registers. Software breakpoints temporarily alter instruction bytes in the private debugged process. Every probed program's output is compared with its uninstrumented counterpart, on both native and ordinary routes. Missing receipts, incomplete returns, event limits, debugger errors and changed observations cannot confirm coverage. Probe failure never replaces an already observed runtime mismatch with an inconclusive result. Execution, output files and event capture are bounded. This is coverage evidence, not a timing or concurrency measurement. Binary and harness hashes are rechecked at the end of the run.

## 6 Current evidence and its limit

The original twelve-family r82 baseline used 266 programs in three storage modes: 798 native/reference pairs. All those observations agree. Another 798 unrelated-definition pairs and 532 cross-storage comparisons agree. All twelve fixed observer controls agree with their expected

outputs on both routes. Fifty-three observer/mutant harness controls pass, and the twenty-one emitted scalar admission assertions pass strict Lean checks.

The normal-promotion history directly records 4,074 rows moving into a relation, followed by shrinking during removal. It retains duplicate occurrences and compares the complete result bags and remaining store. This is activation evidence and a finite behavioral control, not a proof of exact ownership or AtomId-reuse safety. Twelve additional take-over programs cross diagonal and star-shaped binary facts with all six observers. These complement the nested tuple-table case; promotion traces confirm the binary take-over in normal storage mode.

The other eleven candidate families now have direct successful-worker evidence from the pinned probes. Coverage is recorded per case and observer; not every context is expected to enter a worker. A future cohort lacking a family witness still reports `observations-agree-coverage-incomplete` and exits 2 when qualification is requested. Family coverage is not a claim that every internal branch, language profile or concurrent interleaving has been exercised. Arithmetic folds now have witnesses for ordered scan, constant-item runs, join moments and exact bigint continuation. Qualification requires all four subroutes, not merely one successful fold. Bigint evidence must belong to a completed native fold, including when the wrapper's tail call removes its frame name. Exact finite-sum oracles additionally check duplicate contributions, the join multiplier and results above int64. Overflowing moments and non-integer scan inputs exercise ordinary fallback.

Of the 798 case/mode cells, 446 carry successful debugger-return evidence and thirteen carry take-over traces. The other 339 retain observation agreement without a positive activation claim; they include streaming, unsupported-value and disabled-storage contexts. All 744 traced native runs and 744 traced ordinary runs preserve their corresponding uninstrumented observations.

The frozen pre-repair baseline fails ten scalar cases while its constant observer controls pass. This qualifies discrepancy detection and shrinking against a real regression rather than only against synthetic comparator mutations. It does not establish detection of every defect in the other families.

Ten selected archived native mutants are also rejected by generated observations: loose strict-order comparison, wrapped integer addition, dropped filters, incorrect grouped keys, a nonzero absent-group count, dropped listed rows, set-valued list multiplicity, per-hole variable frames, ignored guards and unsound dead-rule coverage. These are existing frozen mutant artifacts, not claimed to be isolated patches over r82. Their ordinary routes agree with the repaired executable in 239 generated comparisons.

Five initially survived the narrower corpus. The added controls expose the missing distinctions: equal operands at a strict-order boundary, a group with zero matches, a guard key already present, a ground fact establishing a kept shape before an open fact, and recursive rules missing required firing directions. Independent finite oracles check designated boundary results in addition to differential agreement.

The frame mutant preserves the answer count while replacing concrete stored values with fresh variables; the post-state observer rejects it. The recursive control has two seeds and two incompletely triggered consequences. Either sequential seed order leaves three facts, while unsound batched admission leaves four. The source observes that order-independent count, so no arbitrary choice of seed order is hidden by the comparator. Three retained failing cases reach local fixed points under the declared reductions and replay without discrepancies on r82; these are not global-minimum claims.

### 6.1 Open compact facts expose a false absence

The added existence family is a distinct native admission route, not a successful return from the joined-query worker. Its positive evidence is `space_match_exists_ground_exact` returning an applicable observation; found and absent are separate values. A refusal must not be counted as an absence. The probe records the applicability output as well as the return value.

Eighteen additional programs cross present, absent and open-stored-fact cases with all six observers. The focused 54-comparison tranche has six discrepancies, all in eager storage mode, with no inconclusive cases. Ordinary matching finds the open fact while the specialized existence test reports absence. A minimal source pattern is:

```
!(bind! &s (new-space set))
!(add-atom &s (seen0 $open))
!(== (collapse (match &s (seen0 b) t)) ())
```

The last answer is true natively and false on the ordinary route. Inserting under that absence guard changes the final store as well. The discrepancy reproduces with only `CETTA_NATIVE_QUERY_REFERENCE` varied; the other three reference switches are not necessary. It is related to a retained failure in the original r82 eager-law log (program 17 in that receipt), not a claim that the historical programme-96 issue has been repaired.

The source inspection and pinned trace identify an incomplete exactness certificate: `space_contains_only_exact_atoms` returns true while a variable-containing fact is kept compactly. The full negative certificate must cover all logical matching candidates, not only ordinary rows. The new `MatchExistenceAdmission` laws and controls formalize this boundary. All six focused failures shrink to one open fact and a ground query under the declared reductions while retaining route evidence. This unresolved native discrepancy blocks an unconditional qualification claim; the earlier positive cohort does not erase it. The complete expanded run contains 284 programs and 852 native/reference comparisons: 846 agree and six differ. Of 568 cross-storage comparisons, 562 agree and the same six eager-mode observations differ. The unrelated definition preserves each route's own result in all 852 metamorphic pairs, but the six cross-route failures remain in those variants. The qualification command exits 1. No timeout or invalid-reference case accounts for these discrepancies.

### 6.2 Coordinate existence composes two storage domains

`space_match_exists_flat_linear_view64` first asks compact storage, then ordinary rows. A compact hit completes the observation. A certified compact miss permits a row-only query. An undecidable compact match settles the relevant head before the row query. The lower-level coordinate helper is therefore qualified only together with its enclosing caller's domain evidence.

Twelve pinned scenarios cross ground, open and unrelated-open facts with plain/eager storage and direct/materialized observation. All return their independent fixed answers on r82 and the successor candidate. Read-only debugger receipts attribute nested results to the same enclosing call. Materialized controls execute no view-existence call. Eight negative harness controls reject missing or late settlement, cross-call evidence, false completion, coordinate refusal treated as absence, and a zero compact count without a completed compact check. The checked `MatchExistencePartition` module supplies the domain laws.

### 6.3 A frozen successor candidate closes the expanded gate

The provenance-backed frozen executable `cetta-codex-scoped-owners-20261101` has identity

d89724427430ecaae39ec4e712a711aa6748cea36c2af6a0b081dd3e5ad9e3c0

identifies itself as `cetta1.5.0-dev(python)` and has ELF build ID `cafb754323b51864872221 77aff18629308032fa`. It retains an open first fact as a plain row when its head later takes a relation; the same minimal existence program then agrees on native and ordinary routes. This is observed binary behavior, not an attribution of the repair or a source-build proof.

The expanded candidate run has 284 programs, 852 native/reference comparisons, 852 unrelated-definition comparisons and 568 cross-storage comparisons. Every comparison agrees. All thirteen generated families have direct activation evidence, the coordinate caller has its separate caller-chain evidence, and all four arithmetic-fold subroutes are covered. The candidate also passes the fixed-seed 200-program eager native-query law.

The same hash governs twelve coordinate scenarios, sixteen retained-reader scenarios, nineteen reached publication controls and all ten archived-mutant checks. The joined handoff validates current harness identities and source anchor hashes, the fourteen-route inventory and the eighteen-module formal manifest. This candidate is qualified evidence; selecting it as the project's agreed successor remains an integration decision.

Its saved source patch applies to recorded HEAD `08e9f948756cb71472c4cb02402d4f7c1f13 e873` in an isolated index. Of 5,963 ledger entries, 5,940 indexed files reproduce byte-for-byte, including all 524 runtime and build paths; the remaining 23 are tests. The completed build/test record, patch and ledger are independently hashed. This is a reconstructible source receipt, not a cryptographic proof that the compiler consumed exactly those bytes.

## 7 Allocation failure and partial publication

A separate pinned debugger driver deliberately injects NULL returns at reviewed allocator call instructions, before those calls allocate anything. It does not alter runtime source or executable files. This mutating child probe is distinct from the observation-only coverage probes. Every fault must be reached exactly once, and stopped processes are not counted as completed terminations. Eight harness controls check those classifications.

Nineteen controls cover nested tuple facts, diagonal binary facts and binary star facts: thirteen recoverable failures preserve observations, three fatal snapshot-copy failures terminate with SIGABRT, and three deliberate source-removal mutants produce a detected bag mismatch. Growth failures exercise representation give-back as well as refusal. Native row-kind receipts establish both an array row and an actual bit row in the binary tests. The model is a single NULL return, not arbitrary allocator, errno or concurrency behavior. The driver records full call-site receipts and checks executable, source and harness identities at completion.

## 8 Retained readers and collection

Sixteen further cases cross nested/flat facts, one/seventeen retained readers, plain/eager relation storage and disabled/eager reclamation. Each reader delivers a prefix before the source facts are removed and replaced; its remaining answers must still complete the original bag. Independent oracles also check the new store and an unrelated live owner. All cases agree with their ordinary routes and those oracles. Keys are computed before insertion rather than supplied as source literals, so later input constants do not incidentally retain every tested value.

The pinned layout probe checks registration, cache materialization, exact root callbacks and deregistration. To bound debugger work, it selects the first collection per universe, active snapshot-context set and cache generation. All 582 callbacks in 70 selected live collections

agree with the snapshot fields, and every active snapshot provider in each selected census is visited. Off-mode cases do not count as collection evidence. The program and expected answers are unchanged by this tracing policy. Eleven unit controls reject missing providers, incomplete probes, inexact callbacks and false collection witnesses. One reached allocation fault during kept-reader provider registration terminates with SIGABRT, not normal refusal.

## 9 Reproducing the qualification

The test entry points live under `tests/support/` in the CeTTa checkout. In the commands below, replace `BINARY` with the executable path and `HASH` with its reviewed SHA-256. The r82 hash is `2acbee5636155b7b175c5965e3fb6d0fbc12b50064f56b39f018a59c2c2d4aed`; the qualified successor-candidate hash is the one above.

```
python3 tests/support/check_native_observers.py BINARY \
  --out COHORT --per-family 6 --modes off on eager \
  --gdb-probe --emit-lean --qualified --expected-sha256 HASH
```

The runner clears inherited `CETTA_*` switches, installs the declared route settings and records relevant runtime settings with each comparison. Specialized drivers explicitly install their collection settings afterward.

The retained-reader and provider-growth controls run separately:

```
python3 tests/support/check_reader_lifetimes.py BINARY --out READERS
python3 tests/support/check_existence_domains.py BINARY --out DOMAINS
python3 tests/support/check_relation_publication_faults.py BINARY \
  --out FAULTS --shape deep
```

Repeat the publication driver with `--shape flat` and `--shape flat-star`, choosing a fresh output directory each time. Unknown executable hashes are refused.

The archived-mutant driver checks individually hash-pinned artifacts, ordinary-route agreement and three minimized replays. Its five gate controls reject survival, timeouts, reference disagreement and unknown statuses. Artifacts have the layout `ARTIFACTS/NAME/mutant`; they are inputs, not binaries rebuilt by the checker:

```
python3 tests/support/check_native_observer_mutants.py BINARY \
  --oracle-run COHORT --mutants-root ARTIFACTS --out MUTANTS
python3 tests/support/check_native_route_inventory.py BINARY --out INVENTORY
python3 tests/support/check_native_source_receipt.py BINARY \
  --repo REPO --patch PATCH --ledger LEDGER --build-log BUILD_LOG \
  --out SOURCE_RECEIPT
```

Every output directory must be new. The emitted scalar Lean file and `RelationCorrespondenceAudit` supply the separate formal checks. Mutation detection can pass on its selected, agreeing oracle cases while another family has a production discrepancy; the mutation receipt retains that full-cohort status. It does not turn the failing native gate green. `check_native_qualification_handoff.py` joins the completed cohort, domain, reader, three publication, mutant and inventory receipts with the Lean foundation-manifest JSON. It exits successfully only when every package names the same unchanged binary, the source receipt is qualified and every component gate is green.

The separate `RelationOwnershipTransfer` note gives the eighteen checked mathematical modules and the remaining native implementation obligations; neither its proofs nor this finite gate certify the whole C runtime.